

$$\begin{aligned} Z_i &= \\ I(X_i \geq \\ x_0) \\ X_i \\ x_0 \\ \{Y_i(1), Y_i(0)\} \\ \tau_Y(x_0) &= \lim_{\varepsilon \rightarrow 0+} E(Y \mid X = x_0 + \varepsilon) - \lim_{\varepsilon \rightarrow 0+} E(Y \mid X = x_0 - \varepsilon). \end{aligned}$$

(1)

$$\begin{aligned} Z_i &= \\ I(X_i \geq \\ x_0) \\ Z_i \\ D_i \\ X_i \geq \\ x_0 \\ Z_i \\ D_i \\ Y_i \\ X_i \\ x_0 \\ Z_i &= I(X_i \geq x_0). \end{aligned}$$

$$\begin{aligned} D_i \\ Z_i \\ P(D_i = \\ 1 \mid \\ X_i = \\ x) \\ x_0 \\ Y_i \\ Z_i \\ \{D_i(1), D_i(0), Y_i(1), Y_i(0)\} \\ \tau_D(x_0) &= E\{D(1) - D(0) \mid X = x_0\} \end{aligned}$$

$$\tau_Y(x_0) = E\{Y(1) - Y(0) \mid X = x_0\}.$$

(2)

$$\begin{aligned} Z \\ D \\ X = \\ x_0 \\ Z \\ I(\bar{X} \geq \\ x_0) \\ x_0 \end{aligned}$$

$$(3) \quad D_i(1) \geq D_i(0),$$

$$(4) \quad D_i(1) = D_i(0) \implies Y_i(1) = Y_i(0),$$

$$\begin{aligned} x_0 \\ E\{D(1) \mid \\ X = \\ x\} \\ E\{Y(1) \mid \\ X = \\ x\} \\ x_0 \\ E\{D(0) \mid \\ X = \\ x\} \\ E\{Y(0) \mid \\ X = \\ x\} \\ x_0 \\ \tau_{\text{LATE}}(x_0) &= E\{Y(1) - Y(0) \mid D(1) > D(0), X = x_0\}, \end{aligned}$$

(5)

$$\tau_{\text{LATE}}(x_0) = \frac{E\{Y(1) - Y(0) \mid X = x_0\}}{E\{D(1) - D(0) \mid X = x_0\}}.$$

(6)

$$\begin{aligned} E(D \mid \\ X = \\ x) \\ x_0 \\ \tau_{\text{LATE}}(x_0) &= \frac{\lim_{\varepsilon \rightarrow 0+} E(Y \mid Z = 1, X = x_0 + \varepsilon) - \lim_{\varepsilon \rightarrow 0+} E(Y \mid Z = 0, X = x_0 - \varepsilon)}{\lim_{\varepsilon \rightarrow 0+} E(D \mid Z = 1, X = x_0 + \varepsilon) - \lim_{\varepsilon \rightarrow 0+} E(D \mid Z = 0, X = x_0 - \varepsilon)}. \end{aligned}$$

(7)

$$\begin{aligned} \text{th:frdd-} \\ \text{identification} \\ Z \\ h \\ x_0 \\ X_i \in \end{aligned}$$